

MTL106: Probability and Stochastic Processes
Lecture 16
February 6, 2026

Correlation: If $E(X^2), E(Y^2)$ exist, we define the correlation coefficient between X and Y as

$$\rho = \frac{\text{Cov}(X, Y)}{\text{SD}(X)\text{SD}(Y)} = \frac{E(XY) - E(X)E(Y)}{\sqrt{E(X^2) - (E(X))^2}\sqrt{E(Y^2) - (E(Y))^2}},$$

where SD stands for the standard deviation. Note that the correlation coefficient ρ is a measure of the linear dependence between X and Y .

Remark 1: We say that random variables X and Y are uncorrelated if $\rho = 0$, or equivalently, $\text{Cov}(X, Y) = 0$.

Remark 2: If X and Y are independent, then $\text{Cov}(X, Y) = 0$ and, X and Y are uncorrelated. If, however, $\rho = 0$, then X and Y may not necessarily be independent.

Example 1: Let U and V be two random variables with common mean and common variance. Let $X = U + V$ and $Y = U - V$. Then

$$\text{Cov}(X, Y) = E(U^2 - V^2) - E(U + V)E(U - V) = 0$$

so that X and Y are uncorrelated but not necessarily independent.

Theorem 1: The correlation coefficient ρ between two random variables X and Y satisfy $|\rho| \leq 1$.

Proof: From Cauchy-Schwarz inequality we have

$$E(XY)^2 \leq E(X^2)E(Y^2)$$

for two random variables X and Y . Now we have

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))].$$

Considering $A = X - E(X)$, $B = Y - E(Y)$, we have

$$\text{Cov}(X, Y) = E[AB].$$

Thus we can write

$$\begin{aligned}
E(AB)^2 &\leq E(A^2)E(B^2). \\
\Rightarrow [Cov(X, Y)]^2 &\leq Var(X)Var(Y). \\
\Rightarrow |Cov(X, Y)| &\leq |SD(X)SD(Y)|. \\
\Rightarrow \left| \frac{Cov(X, Y)}{SD(X)SD(Y)} \right| &\leq 1. \\
\Rightarrow |\rho| &\leq 1.
\end{aligned}$$

Theorem 2: Let $E(X^2) < \infty$, $E(Y^2) < \infty$, and let $U = aX + b$, $V = cY + d$. Then

$$\rho_{X,Y} = \pm \rho_{U,V}.$$

Proof: We have

$$\begin{aligned}
\rho_{X,Y} &= \frac{\pm ac E \left[\left\{ \frac{U-b}{a} - E \left(\frac{U-b}{a} \right) \right\} \left\{ \frac{V-d}{c} - E \left(\frac{V-d}{c} \right) \right\} \right]}{\sigma_U \sigma_V} \\
&= \frac{\pm \{E(UV) - E(U)E(V)\}}{\sigma_U \sigma_V} \\
&= \pm \rho_{U,V}.
\end{aligned}$$

Theorem 3: Let $X_1, X_2, \dots, X_n$ be random variables such that $E|X_i| < \infty$, $i = 1, 2, \dots, n$. Let $a_1, a_2, \dots, a_n$ be real numbers, and write

$$S = \sum_{j=1}^n a_j X_j.$$

Then $E(S)$ exists, and we have

$$E(S) = \sum_{j=1}^n a_j E(X_j). \quad (1)$$

Corollary: Take $a_1 = a_2 = \dots = a_n = 1/n$. Then

$$E \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right) = \frac{1}{n} \sum_{i=1}^n E(X_i),$$

and if $E(X_1) = E(X_2) = \dots = E(X_n) = \mu$, then

$$E \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right) = \mu.$$

Some Discrete Probability Distributions

Bernoulli Distribution

- Any random experiment whose outcomes are of two types, success (S) and failure (F), occurring with probabilities p and q respectively, is called a **Bernoulli Trial**.
- Let $(\Omega, \mathcal{F}, P)$ be a probability space associated with an experiment, and let $A \in \mathcal{F}$ with $P(A) = p > 0$. Then $P(A^c) = 1 - p = q$. Each performance of the experiment is a Bernoulli Trial. It will be convenient to call the occurrence of event A a success and the occurrence of A^c a failure.

Bernoulli Distribution (Contd.)

Applications:

- In a manufacturing plant, the quality control department may inspect items on a production line. Each item is either defective (failure) or non-defective (success).
- In the insurance industry, each policyholder might file a claim (success) or not file a claim (failure) within a certain period.
- In medical testing, a diagnostic test might yield either a positive result (presence of a disease) or a negative result (absence of a disease). The result of each test can be modeled as a Bernoulli trial, with success corresponding to a positive result.

Bernoulli Distribution (Contd.)

A random variable X is said to follow a **Bernoulli distribution** with parameter p , if its probability mass function (PMF) is given by

$$P\{X = x\} = \begin{cases} p^x(1-p)^{1-x}, & x = 0, 1, \\ 0, & \text{otherwise,} \end{cases}$$

where $p \in [0, 1]$. Here X is defined such that it takes value 1 when success (S) occurs and 0 when failure (F) occurs.

Bernoulli Distribution (Contd.)

- **Mean:** $E(X) = 1 \cdot p + 0 \cdot (1 - p) = p.$
- **Second Moment:** $E(X^2) = 1^2 \cdot p + 0^2 \cdot (1 - p) = p.$
- **Variance:** $\text{Var}(X) = E(X^2) - (E(X))^2 = p - p^2 = p(1 - p) = pq.$
- **MGF:** $M_X(t) = E(e^{tX}) = q + pe^t.$

Uniform Distribution

A random variable X is said to have a **uniform distribution** on n points $\{x_1, x_2, \dots, x_n\}$ if its PMF is of the form

$$P\{X = x_i\} = \frac{1}{n}, \quad i = 1, 2, \dots, n.$$

Applications:

- In a lottery where a number is drawn randomly from a set of 50 numbers $(1, 2, 3, \dots, 50)$, each number has an equal chance of being selected. The selection of the winning number can be modeled by a discrete uniform distribution.
- In a survey where a random sample is drawn from a population, and each member of the population has an equal chance of being selected, the selection process can be modeled by a discrete uniform distribution.

Uniform Distribution (Contd.)

Mean:

$$E(X) = \sum_{i=1}^n x_i P(X = x_i) = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}.$$

Second Moment:

$$E(X^2) = \sum_{i=1}^n x_i^2 P(X = x_i) = \frac{1}{n} \sum_{i=1}^n x_i^2.$$

Variance:

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

If $x_i = i$, $i = 1, 2, \dots, n$, then:

$$E(X) = \frac{n+1}{2}, \quad E(X^2) = \frac{(n+1)(2n+1)}{6}, \quad \text{Var}(X) = \frac{n^2-1}{12}.$$

Binomial Distribution

- Consider a series of n independent Bernoulli trials (where n is finite) in which the constant probabilities of success and failure in any trial are p and $q = 1 - p$, respectively.
- Let X denote the number of successes. The x successes in n trials can occur in $\binom{n}{x}$ ways.
- Thus, the probability of x successes [or $(n - x)$ failures] in n trials is given by

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n; \quad 0 \leq p \leq 1.$$

- We say that X has a **Binomial distribution** with parameter p [$X \sim b(n, p)$].

Binomial Distribution (Contd.)

Applications:

- Any given context in which there are only two outcomes can be analyzed using Binomial distribution.
- Customer churn: churn vs no churn.
- Insurance claims: fraud vs genuine.
- Examination result: pass vs fail.
- Quality check: defective vs non-defective.

Binomial Distribution (Contd.)

$$\sum_{x=0}^n \binom{n}{x} p^x (1-p)^{n-x} = 1.$$

- **Mean:** $E(X) = np$.
- **Second Moment:** $E(X^2) = n(n-1)p^2 + np$.
- **Variance:** $\text{Var}(X) = np(1-p) = npq$.
- **MGF:** $M_X(t) = (q + pe^t)^n$.
- **PGF:** $P(s) = (q + ps)^n, |s| \leq 1$.

Binomial Distribution (Contd.)

Theorem 1 (Additive Property). Let $X_i \sim \text{Binomial}(n_i, p)$ ($i = 1, 2, \dots, k$) be independent random variables. Then

$$S_k = \sum_{i=1}^k X_i \sim \text{Binomial} \left(\sum_{i=1}^k n_i, p \right).$$

Proof: Since $X_1, \dots, X_k$ are independent, the MGF of $S_k = \sum_{i=1}^k X_i$ can be written as

$$\begin{aligned} M_{S_k}(t) &= \prod_{i=1}^k M_{X_i}(t) \\ &= (q + pe^t)^{n_1} (q + pe^t)^{n_2} \cdots (q + pe^t)^{n_k} \\ &= (q + pe^t)^{\sum_{i=1}^k n_i}. \end{aligned}$$

Hence, $S_k \sim \text{Binomial} \left(\sum_{i=1}^k n_i, p \right)$ (using the uniqueness property of MGF).

Binomial Distribution (Contd.)

R Commands:

- `dbinom(x, size, prob)`: $P(X = x)$.
- `pbinom(q, size, prob)`: $P(X \leq q)$.
- `qbinom(p, size, prob)`: smallest q such that $P(X \leq q) \geq p$.
- `rbinom(n, size, prob)`: generates n random observations.

Poisson Distribution

In many circumstances, one might be interested in knowing the number of events that may occur over a period of time (or corresponding unit of measurement).

Poisson Distribution (Contd.)

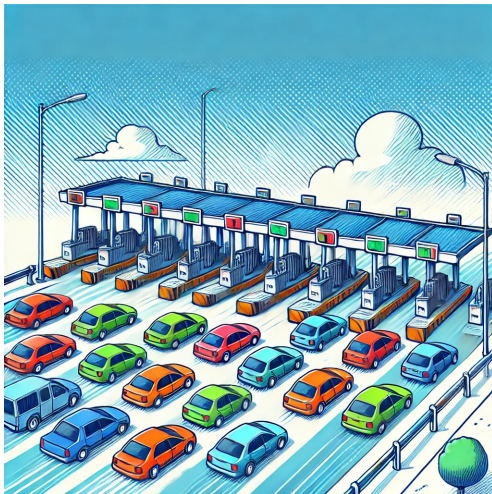


Figure 1: Cars passing through a tollbooth (Google image)

Poisson Distribution (Contd.)



Figure 2: Potholes on a road (Google image)

Poisson Distribution (Contd.)



Figure 3: Cash withdrawals in an ATM (Google image)

Poisson Distribution (Contd.)



Figure 4: Customer arrivals at a store (Google image)

Poisson Distribution (Contd.)



Figure 5: Phone calls receiving at a call center (Google image)

Poisson Distribution (Contd.)

The probability of number of events occurring within a fixed time interval (or corresponding unit of measurement) can be modelled using Poisson distribution.

Poisson Distribution (Contd.)

- A random variable X is said to be a **Poisson** random variable with parameter $\lambda > 0$ if its **PMF** is given by

$$P\{X = k\} = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots \quad (1)$$

- The parameter λ in (1) is known as the *rate of occurrence* of the events per unit of measurement (time).
- Equation (1) indeed defines a PMF. We have

$$\sum_{k=0}^{\infty} P\{X = k\} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1.$$

- If X has the PMF given by (1), we will write $X \sim P(\lambda)$.

Poisson Distribution (Contd.)

- **CDF:** $F(x) = P\{X \leq k\} = \sum_{i=0}^k P\{X = i\} = e^{-\lambda} \sum_{i=0}^k \frac{\lambda^i}{i!}.$
- **Mean:** $E(X) = e^{-\lambda} \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda.$
- **Second Moment:** $E(X^2) = e^{-\lambda} \sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!} =$
 $e^{-\lambda} \sum_{k=0}^{\infty} [k(k-1) + k] \frac{\lambda^k}{k!} = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} + \lambda = \lambda + \lambda^2.$
- **Variance:** $Var(X) = E(X^2) - [E(X)]^2 = \lambda.$
- **MGF:** $M_X(t) = e^{-\lambda} \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k}{k!} = e^{\lambda(e^t-1)}.$
- **PGF:** $P_X(s) = e^{-\lambda} \sum_{k=0}^{\infty} s^k \frac{\lambda^k}{k!} = e^{-\lambda(1-s)}, \quad |s| \leq 1.$

Poisson Distribution (Contd.)

If we are interested in predicting the number of events over a period of time, say between 0 and t , then the PMF is given by

$$P\{N(t) = k\} = \frac{e^{-\lambda t}(\lambda t)^k}{k!},$$

where $N(t)$ is the number of events that occur over a period of time t or in $[0, t]$. The parameter λ is the mean number of occurrences per unit time and λt is the mean number of occurrences in an interval $[0, t]$. We also assume that the events are independent.

Poisson Distribution (Contd.)

Theorem 1 (Additive Property)

Let $X_1, X_2, \dots, X_n$ be independent Poisson random variables with $X_k \sim P(\lambda_k), k = 1, 2, \dots, n$. Then $S_n = \sum_{k=1}^n X_k$ is a $P\left(\sum_{k=1}^n \lambda_k\right)$ random variable.

Proof: Since $X_1, X_2, \dots, X_n$ are independent, the MGF of $S_n = \sum_{k=1}^n X_k$ can be written as

$$\begin{aligned} M_{S_n}(t) &= M_{X_1}(t)M_{X_2}(t)\dots M_{X_n}(t) \\ &= e^{\lambda_1(e^t-1)}e^{\lambda_2(e^t-1)}\dots e^{\lambda_n(e^t-1)} \\ &= e^{(\lambda_1+\lambda_2+\dots+\lambda_n)(e^t-1)}. \end{aligned}$$

$$\Rightarrow S_n \sim P\left(\sum_{k=1}^n \lambda_k\right) \text{ (using the uniqueness property of MGF).}$$

Poisson Distribution (Contd.)

Theorem 2

Let X and Y be independent Poisson random variables with parameters λ_1 and λ_2 respectively. Then the conditional distribution of X , given $X + Y$, is binomial.

Proof: For integers $m > 0, n > 0 (m < n)$, we have

$$\begin{aligned} P\{X = m | X + Y = n\} &= \frac{P\{X = m, Y = n - m\}}{P\{X + Y = n\}} \\ &= \frac{\frac{e^{-\lambda_1} \lambda_1^m}{m!} \frac{e^{-\lambda_2} \lambda_2^{n-m}}{(n-m)!}}{\frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^n}{n!}} \\ &= \binom{n}{m} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^m \left(1 - \frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^{n-m}, \end{aligned}$$

where $m = 0, 1, 2, \dots, n$, and the proof is complete.

Poisson Distribution (Contd.)

R Commands:

- `dpois(x, lambda)` gives the probability of observing exactly x "events" in an interval, where "lambda" is the average rate of occurrence (mean) in an interval.
- `ppois(q, lambda)` gives the cumulative probability of observing q or fewer events.
- `qpois(p, lambda)` gives the "quantile function", i.e., the smallest value q such that the cumulative probability is at least p .
- `rpois(n, lambda)` generates n random observations from the Poisson distribution.